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Q1. OSI Reference Model

Marks: 10M | ■ Every Year –
GUARANTEED

Q1.

[10M | ■
Every
Year]

OSI Reference Model explain karo — 7 layers, unke functions, protocols aur services diagram ke saath. TCP/IP model se compare karo.

Answer:

OSI (Open Systems Interconnection) model ISO (International Standards Organization) dwara 1984 mein develop kiya gaya ek **7-layer reference model** hai. Ye define karta hai ki do different systems ek doosre se kaise communicate kar sakte hain — chahe unka hardware ya software alag kyon na ho. Ye ek conceptual framework hai, actual implementation nahi.

Layer #	Layer Name	Key Functions	Key Protocols/Examples
7	Application	User interface, Network services access,	HTTP, FTP, SMTP, DNS, Telnet, SNMP
6	Presentation	Data translation, Encryption/Decryption, Compression, Format Conversion,	SSL/TLS, JPEG, MPEG, ASCII, EBCDIC
5	Session	Session setup/maintain/terminate, Synchronization, Dead-end Control	NetBIOS, RPC, PPTP
4	Transport	End-to-end delivery, Segmentation, Flow Control, Port addressing	TCP, UDP, SCTP
3	Network	Logical addressing, Routing, Packet forwarding,	IP, ICMP, OSPF, BGP, ARP
2	Data Link	Framing, MAC addressing, Error detection (CRC), Flow control, Access (802.11)	Ethernet, PPP, HDLC, Access (802.11)
1	Physical	Bit transmission, Cables, Connectors, Voltage levels, Data rate, Fiber optic, RS-232	Ethernet, Data Bus, Fiber

Data Flow in OSI Model (Sender → Receiver):

Sender side pe data Application layer se Physical layer tak jaata hai — har layer apna **header (aur kabhi trailer) add** karti hai (encapsulation). Receiver side pe Physical layer se Application layer tak jaata hai — har layer apna header remove karti hai (decapsulation).

Application → Presentation → Session → Transport → Network → Data Link → Physical
(Data) (Data) (Data) (Segment) (Packet) (Frame) (Bits)

OSI vs TCP/IP Model Comparison:

Feature	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Layers	7 layers	4 layers
Layer names	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network, Transport, Internet, Application
Development	By ISO (1984) – Theoretical	By DARPA (1970s) – Practical
Protocol specific?	Not tied to any protocol	Based on TCP/IP suite
Flexibility	More rigid	More flexible

Usage	Reference/teaching model	Used in real Internet
Session layer	Separate (Layer 5)	Part of Application layer
Presentation layer	Separate (Layer 6)	Part of Application layer
Reliability	Defined in Transport & Data Link	Mainly in Transport (TCP)

TCP/IP 4 Layers:

1. **Network Access Layer** (= Physical + Data Link of OSI) — Ethernet, Wi-Fi
2. **Internet Layer** (= Network of OSI) — IP, ICMP, ARP
3. **Transport Layer** (= Transport of OSI) — TCP, UDP
4. **Application Layer** (= Session + Presentation + Application of OSI) — HTTP, FTP, SMTP, DNS

Q2. Bit Rate vs Baud Rate + Nyquist + Shannon

Marks: 2-5M | ■ Repeat

Q2.

[2-5M | ■ Repeat]

Bit Rate aur Baud Rate mein kya difference hai? Nyquist formula aur Shannon theorem explain karo. Numerical: Channel capacity calculate karo.

Bit Rate vs Baud Rate:

Feature	Bit Rate	Baud Rate
Definition	Bits transmitted per second	Signal changes (symbols) per second
Unit	bps (bits per second)	Baud
Measures	Data throughput	Signaling rate
Relationship	Bit Rate = Baud Rate $\times \log_2(M)$	Baud Rate = Bit Rate / $\log_2(M)$
When equal?	When $M=2$ (binary signaling)	Same as bit rate only for binary
Example	4800 bps	1200 baud (with 4 signal levels)

Where **M** = number of signal levels (symbols). Example: If $M=4$ (4 levels like 00,01,10,11) aur Baud Rate=1200, then Bit Rate = $1200 \times \log_2(4) = 1200 \times 2 = \mathbf{2400 \text{ bps}}$.

Nyquist Theorem (Noiseless Channel):

Nyquist theorem bolta hai ki ek noiseless channel mein maximum data rate:

$$\text{Max Bit Rate} = 2 \times B \times \log_2(M) \text{ bits/sec}$$

Where: **B** = Bandwidth (Hz), **M** = number of discrete signal levels

Numerical: Bandwidth $B = 3000 \text{ Hz}$, Signal levels $M = 8$

$$\text{Max Bit Rate} = 2 \times 3000 \times \log_2(8) = 2 \times 3000 \times 3 = \mathbf{18,000 \text{ bps} = 18 \text{ kbps}}$$

Shannon's Theorem (Noisy Channel):

Shannon's theorem bolta hai ki ek noisy channel mein maximum theoretical capacity:

$$C = B \times \log_2(1 + \text{SNR}) \text{ bits/sec}$$

Where: **B** = Bandwidth (Hz), **SNR** = Signal-to-Noise Ratio (dimensionless ratio)

$$\text{SNR (dB)} = 10 \times \log_{10}(\text{SNR}) \rightarrow \text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$$

Numerical: Telephone line: $B = 3000 \text{ Hz}$, $\text{SNR} = 30 \text{ dB}$

First convert: $\text{SNR (linear)} = 10^{(30/10)} = 10^3 = 1000$

$$C = 3000 \times \log_2(1 + 1000) = 3000 \times \log_2(1001) \approx 3000 \times 9.97 \approx \mathbf{29,900 \text{ bps} \approx 30 \text{ kbps}}$$

Key Difference: Nyquist = noiseless channel, depends on bandwidth + signal levels. Shannon = noisy channel, depends on bandwidth + SNR. **Actual maximum = min(Nyquist limit, Shannon limit).**

Q3. Multiplexing – FDM, TDM, WDM, STDM

Marks: 10M | ■ Repeat

Q3.

[10M | ■ Repeat]

Multiplexing kya hai aur uske types explain karo: FDM, TDM, WDM, STDM — working, diagram aur differences ke saath.

Multiplexing kya hai?

Multiplexing ek technique hai jisme **multiple signals ko ek hi shared medium/channel** pe simultaneously transmit kiya jaata hai. Isse bandwidth efficiently use hoti hai aur cost kam hoti hai. Receiver side pe demultiplexer individual signals alag karta hai.

1. FDM – Frequency Division Multiplexing

Available bandwidth ko multiple frequency bands mein divide kiya jaata hai. Har signal/user ko ek alag frequency band allocate hoti hai. Ye bands simultaneously transmit hoti hain.

Use case:	FM/AM Radio, Cable TV, Traditional telephone
Guard bands:	Interference rokne ke liye bands ke beech chhoti gap hoti hai
Type:	Analog multiplexing technique
Drawback:	Bandwidth waste hoti hai guard bands ki wajah se

2. TDM – Time Division Multiplexing

Available time ko slots mein divide kiya jaata hai. Har user/signal ko ek time slot milta hai — cyclically. Ek complete cycle = ek frame.

Types:	Synchronous TDM (fixed slots) aur Statistical TDM (dynamic slots)
Use case:	Digital telephony (T1/E1 lines), SONET/SDH
Efficiency issue:	Synchronous TDM mein idle slots waste hote hain
Type:	Digital multiplexing technique

3. WDM – Wavelength Division Multiplexing

Optical fiber mein different wavelengths (colors) of light use karke multiple signals simultaneously transmit kiye jaate hain. Essentially FDM but for optical signals.

Types:	DWDM (Dense WDM – 80+ channels) aur CWDM (Coarse WDM – 8-16 channels)
Use case:	Long-distance fiber optic communications, Internet backbone
Advantage:	Extremely high bandwidth (Tbps range)

Medium: Only optical fiber

4. STDM – Statistical TDM (Asynchronous TDM)

Synchronous TDM ka improved version. Time slots dynamically allocate hote hain sirf un users ko jo data transmit karna chahte hain. Idle users ko slot nahi milti.

Advantage: No wasted slots — better efficiency than synchronous TDM

Drawback: Header information add karni padti hai (which user ka data hai)

Use case: Statistical multiplexing in packet networks, ATM

Also called: Asynchronous TDM or Intelligent TDM

Comparison Table:

Feature	FDM	TDM	WDM	STDM
Signal type	Analog	Digital	Optical	Digital
Division by	Frequency	Time	Wavelength	Time (dynamic)
Medium	Any	Any	Fiber only	Any
Guard bands?	Yes	No	Yes (guard λ)	No
Efficiency	Medium	Lower (idle waste)	Very High	Higher
Cost	Medium	Low	High	Medium

Q4. Circuit Switching vs Packet Switching

Marks: 10M | ■ Repeat

Q4.

[10M | ■ Repeat]

Circuit Switching aur Packet Switching mein kya difference hai? Advantages, disadvantages aur use cases ke saath compare karo. Message Switching bhi include karo.

Circuit Switching:

Circuit Switching mein communication shuru hone se pehle source aur destination ke beech ek **dedicated physical path (circuit)** establish hoti hai. Ye path poori communication ke dauran reserved rehti hai — chahe data ja raha ho ya nahi. Traditional telephone network (PSTN) ka example hai.

Circuit Switching ke 3 Phases:

- **Connection Setup:** Source aur destination ke beech dedicated circuit establish hoti hai (resources reserve ho jaate hain).
- **Data Transfer:** Data continuously dedicated circuit pe travel karta hai (guaranteed bandwidth).
- **Connection Teardown:** Communication khatam hone ke baad circuit release hoti hai aur resources free hote hain.

Packet Switching:

Packet Switching mein data ko chhote **packets** mein tod diya jaata hai. Har packet independently route hota hai network mein — alag paths le sakta hai. Packets destination pe reassemble hote hain. Koi dedicated path nahi hoti — resources shared hain (statistical multiplexing). Internet ka basis hai.

Packet Switching ke 2 Types:

Datagram (Connectionless): Har packet independently route hota hai. Packets order out-of-order pahunch sakte hain. Example: IP protocol.

Virtual Circuit (Connection-oriented): Pehle virtual path establish hoti hai, phir sabhi packets same path follow karte hain. Example: ATM, Frame Relay.

Message Switching:

Message Switching mein poora message (bina todhe) store-and-forward technique se intermediate nodes pe store hota hai. Har node puri message store karta hai phir next node pe forward karta hai. Aaj practically obsolete hai.

Comparison Table:

Feature	Circuit Switching	Packet Switching	Message Switching
Path	Dedicated path	No dedicated path	Store and forward
Bandwidth	Reserved	Shared (on demand)	Variable
Delay	Low (after setup)	Variable (queuing)	High
Setup required?	Yes	No (datagram)	No
Resource waste	Yes (idle circuits)	No waste	Depends

Failure handling	Call drops	Re-routing possible	Delayed delivery
Best for	Real-time voice	Data networks	Email, telegram (old)
Example	PSTN, ISDN	Internet (IP)	Old telegram systems
Cost	Higher	Lower	Medium

Advantages of Packet Switching over Circuit Switching:

- Better bandwidth utilization (no idle reservation)
- Multiple users share same network simultaneously
- More fault tolerant — packets can take alternate routes
- Better for bursty data traffic (internet browsing, file transfer)

Q5. Guided Transmission Media

Marks: 10M | High

Q5.

[10M | ■
High]

Guided transmission media explain karo — Twisted pair, Coaxial cable, Fiber optic. Unke types, properties, advantages aur disadvantages.

Guided media mein signal ek physical wire ya cable ke andar travel karta hai. Signal ka direction wire se guide hota hai (isliye "guided"). Teeno main types hain:

1. Twisted Pair Cable

Do insulated copper wires ko ek doosre ke around twist kiya jaata hai. Twisting electromagnetic interference (EMI) reduce karta hai.

UTP (Unshielded Twisted Pair):	No shielding. Cheapest. Categories: Cat3 (10 Mbps), Cat5 (100 Mbps), Cat5e (1 Gbps), Cat6 (10 Gbps). LAN mein most common.
STP (Shielded Twisted Pair):	Metal foil shield covers the pairs. Better EMI protection. More expensive. Used in industrial environments.
Bandwidth:	100 MHz to 250 MHz depending on category. Distance up to 100m.
Applications:	Ethernet LAN, Telephone networks, DSL internet.
Advantage:	Cheap, easy to install, flexible.
Disadvantage:	Susceptible to interference over long distance, attenuation.

2. Coaxial Cable

Inner copper conductor ke around insulation, braided metal shield, aur outer jacket hoti hai. Central conductor signal carry karta hai, shield ground aur interference protection provide karta hai.

Types:	Thicknet (10Base5, 500m) aur Thinnet (10Base2, 185m). Also RG-6, RG-11 for TV.
Bandwidth:	100 MHz to 500 MHz. Up to 10 Gbps in some applications.
Applications:	Cable TV (CATV), Cable internet, Antenna connections, Old Ethernet.
Advantage:	Better shielding than twisted pair, higher bandwidth.
Disadvantage:	More expensive, thicker/heavier, harder to install.

3. Fiber Optic Cable

Glass ya plastic fiber ke through light pulses transmit hote hain. Total internal reflection ka principle use hota hai. Highest bandwidth, lowest attenuation.

Single-mode fiber:	Very thin core (8-10 μ m). Long distance (100+ km). Uses laser. For WAN, telecom backbone.
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Multi-mode fiber:	Thicker core (50-62.5 μm). Shorter distance (<2 km). Uses LED. For LAN, data centers.
Bandwidth:	Terabits per second — virtually unlimited.
Applications:	Internet backbone, WAN, undersea cables, Long-haul telecom.
Advantage:	Highest bandwidth, immune to EMI, very low attenuation, secure (no signal leakage).
Disadvantage:	Expensive, fragile, difficult to splice/install, one-directional per fiber.

Comparison Table – Guided Media:

Feature	Twisted Pair	Coaxial Cable	Fiber Optic
Medium	Copper wire	Copper + shield	Glass/Plastic fiber
Signal type	Electrical	Electrical	Light (optical)
Bandwidth	Up to 250 MHz	Up to 500 MHz	Terabits/sec
Distance	Up to 100m	Up to 500m	Up to 100 km+
Cost	Cheapest	Medium	Most expensive
EMI immunity	Low	Medium	Complete immunity
Security	Low	Medium	Very high
Installation	Easy	Medium	Difficult
Use case	LAN, Telephone	Cable TV, CATV	Backbone, WAN

Q6. Unguided Transmission Media

Marks: 5M | High

Q6. [5M]
[■ High]

Unguided (Wireless) transmission media explain karo — Radio waves, Microwave, Infrared. Properties, frequency range aur applications.

Unguided media mein signal free space mein electromagnetic waves ke roop mein travel karta hai — koi physical cable nahi hoti. Signal "broadcast" hota hai.

Type	Freq Range	Propagation	Applications	Key Properties
Radio Waves	3 Hz – 1 GHz	Omnidirectional, penetrates walls	AM/FM Radio, TV broadcast, Wi-Fi (2.4 GHz)	Low frequency, unregulated spectrum
Microwave	1 GHz – 300 GHz	Line-of-sight (LOS)	Satellite comm, Cellular networks, Radar	High frequency, LOS, dish antennas
Infrared	300 GHz – 400 THz	Very short range, no walls	Remote, IrDA (device-to-device), Li-Fi	Short range, LOS, very short range
Satellite	500 MHz – 2.7 GHz	Earth orbit to ground	GPS, Satellite TV, Internet	High latency (e.g. ~270ms), wide coverage

Guided vs Unguided – Key Difference:

- Guided: Physical medium required (wire/cable). More secure, less interference. Less mobile.
- Unguided: No physical medium. Wireless, mobile, broadcast possible. More interference susceptible.

Q7. Modulation – Analog + Digital

Marks: 5-10M | High

Q7.

[10M | High]

Modulation kya hai? Types explain karo: AM, FM, PM (Analog) aur ASK, FSK, PSK, QAM (Digital).

Modulation kya hai?

Modulation ek process hai jisme data signal (message) ko ek higher-frequency **carrier signal** ke saath combine kiya jaata hai transmission ke liye. Carrier signal ki koi ek ya zyada characteristics (amplitude, frequency, phase) ko data ke anusaar modify kiya jaata hai.

Carrier signal: $C(t) = A \times \sin(2\pi f t + \phi)$ Modulate karte hain: A (amplitude), f (frequency), ϕ (phase)

Part A – Analog Modulation (Analog data on Analog carrier):

Type	Full Name	What changes	Bandwidth	Use case	Key property
AM	Amplitude Modulation	Carrier amplitude varies with signal	2 × signal bandwidth	AM radio (540 kHz – 1.6 MHz)	Sensitive to noise
FM	Frequency Modulation	Carrier frequency varies with signal	5 × signal bandwidth	FM radio (88–108 MHz)	Better noise immunity than AM
PM	Phase Modulation	Carrier phase varies with signal	Similar to FM	Data modems, some digital systems	Resistant to amplitude variations

Part B – Digital Modulation (Digital data on Analog carrier):

Digital data (bits) ko analog carrier waves par encode karna hota hai — mainly wireless transmission ke liye.

Type	Full Name	Working	Bits/Symbol	Use case
ASK	Amplitude Shift Keying	0 → low/no amplitude; 1 → high amplitude	1	Simple optical links, RFID
FSK	Frequency Shift Keying	0 → frequency f1; 1 → frequency f2	2	Caller ID, early modems, telemetry
PSK	Phase Shift Keying	Phase changes to represent bits	2 (BPSK), 4 (QPSK), 8 (8-PSK), 16 (16-PSK)	Modems, satellite, Wi-Fi (BPSK)
QAM	Quadrature Amplitude Modulation	Combines ASK + PSK. 16-QAM, 64-QAM, 256-QAM	4, 6, 8 bits per symbol	Modern Wi-Fi 802.11ac, 4G

QAM kyon best hai?

QAM ek saath amplitude aur phase dono vary karta hai → zyada symbols possible → zyada bits per symbol → higher throughput at same bandwidth. 256-QAM = 8 bits per symbol, isliye modern Wi-Fi aur 4G mein use hota hai.

Q8. NRZ-L, NRZ-I Encoding – Diagram

Marks: 5M | PYQ 2022

Q8. [5M]
[PYQ 2022]

NRZ-L, NRZ-I, Manchester, Differential Manchester encoding schemes explain karo. Data 01001110 ke liye NRZ-L aur NRZ-I diagram draw karo.

Line Coding kya hai?

Line coding digital data ko digital signals mein convert karne ki process hai. Voltage levels use karke bits represent kiye jaate hain.

Encoding Scheme	Working Rule	Bit 0	Bit 1	Drawback
NRZ-L (Non-Return to Zero Level)	Voltage level directly represents bit value	High voltage (+V)	Low voltage (0V)	Long strings of same bit → sync problems
NRZ-I (Non-Return to Zero Inverted)	Transition at beginning of bit interval represents bit value	Low voltage (0V)	High voltage (+V)	Long strings of 0s still cause sync issues
Manchester	Each bit has a mid-bit transition: High → Low for 0, Low → High for 1	High → Low	Low → High	Requires 2× bandwidth of NRZ
Differential Manchester	Mid-bit transition always present. Transition at start of bit interval represents bit value	Transition at start	No transition at start	More complex to implement

NRZ-L Diagram for Data: 0 1 0 0 1 1 1 0

Rule: 0 = High (+V), 1 = Low (0V) Data: 0 1 0 0 1 1 1 0 Signal: +V | 0V | +V | +V | 0V | 0V | 0V | +V
Diagram (text representation): _____ ||| | _____

NRZ-I Diagram for Data: 0 1 0 0 1 1 1 0

Rule: 1 = transition at start of bit; 0 = no transition Start: assume signal is at Low (0V) Data: 0 1 0 0 1 1 1 0
Action: Stay | Trans | Stay | Stay | Trans | Trans | Trans | Stay Level: 0V | +V | +V | +V | 0V | +V | 0V | 0V
Diagram (text representation): _____ ||| | ||| | _____ ||
| _____

EXAM TIP

Exam Tip: NRZ-L mein signal level → bit value. NRZ-I mein transition presence → bit 1. Manchester mein mid-bit transition hamesha hoti hai — ye synchronization problem solve karta hai. Ethernet (10 Mbps) Manchester encoding use karta hai.

Q9. Network Topologies + Point-to-Point vs Multipoint

Marks: 5M | ■ 2023 PYQ

Q9. [5M]
■ 2023
PYQ]

Point-to-Point aur Point-to-Multipoint mein difference explain karo. Network topologies ke advantages/disadvantages bhi explain karo.

Point-to-Point vs Point-to-Multipoint:

Feature	Point-to-Point	Point-to-Multipoint
Connection	Direct link between 2 devices only	One sender, multiple receivers share link
Dedicated?	Dedicated channel	Shared channel
Cost	Higher (more cables)	Lower (shared medium)
Security	More secure	Less secure (shared)
Example	Leased lines, direct Ethernet, PPP	Bus topology, Wi-Fi broadcast, Cable TV
Bandwidth	Full bandwidth dedicated	Bandwidth shared among users

Network Topologies – Types, Advantages aur Disadvantages:

Topology	Structure	Advantages	Disadvantages	Use Case
Bus	All devices on single cable (backbone), easy to extend	Simple, cheap	Single cable failure = whole network down; Limited length	Old Ethernet (10Base2)
Star	All devices connected to central hub/switch	Easy fault isolation, easy to add/delete devices	Hub/switch failure affects all devices; Requires more cable	Most modern LANs, Wi-Fi
Ring	Devices connected in a closed loop	Equal access, no collisions, predictable performance	Single cable failure = network down; Slow to reconfigure	Token Ring, FDDI (old)
Mesh	Every device connected to multiple others	Highly fault tolerant, multiple paths	Very expensive, complex to install, difficult to manage	WANs, Internet backbone
Tree	Hierarchical star — root switch connects to other switches	Scalable, easy to manage	Root switch failure affects all; More complex	Corporate networks, ISP networks
Hybrid	Combination of 2+ topologies	Flexible, scalable, inherits best properties	Complex, expensive	Large enterprise networks

Quick Revision – Unit 1 Formula & Key Points Sheet

Exam se pehle zaroor padhna

Topic	Key Formula / Key Point
Bit Rate vs Baud Rate	Bit Rate = Baud Rate $\times \log_2(M)$; where M = signal levels
Nyquist Theorem	Max Bit Rate = $2 \times B \times \log_2(M)$ [noiseless channel]
Shannon's Theorem	$C = B \times \log_2(1 + \text{SNR})$ [noisy channel; B=bandwidth]
SNR conversion	$\text{SNR}(\text{linear}) = 10^{(\text{SNR}_{\text{dB}} / 10)}$
OSI Layer 1	Physical - bits, cables, voltage, timing
OSI Layer 2	Data Link - frames, MAC addr, CRC, CSMA/CD
OSI Layer 3	Network - packets, IP, routing, fragmentation
OSI Layer 4	Transport - segments, TCP/UDP, ports, flow control
OSI Layers 5-7	Session, Presentation, Application (merged in TCP/IP)
TCP/IP = 4 layers	Network Access Internet Transport Application
OSI vs TCP/IP	OSI=7 layers (theoretical); TCP/IP=4 layers (practical, Internet)
FDM	Frequency division; Analog; Guard bands; Radio/TV
TDM	Time division; Digital; Fixed slots; Telephone T1/E1
WDM	Wavelength division; Optical fiber only; DWDM=80+ channels
STDM	Statistical/Asynchronous TDM; Dynamic slots; No idle waste
Circuit Switching	Dedicated path; 3 phases; PSTN; Guaranteed BW
Packet Switching	No dedicated path; Packets routed independently; Internet
Twisted Pair	Cheapest; Up to 100m; Cat5e=1Gbps; LAN/Telephone
Coaxial	Metal shield; Up to 500m; Cable TV/CATV
Fiber Optic	Light signal; Immune to EMI; TBps; Long distance
NRZ-L	0=High, 1=Low (or opposite); Level-based
NRZ-I	1=transition at start, 0=no transition; Inversion-based
Manchester	Mid-bit transition always; 0=H→L, 1=L→H; Ethernet
AM	Amplitude changes with signal; AM Radio
FM	Frequency changes with signal; FM Radio; Better noise immunity
ASK	Amplitude Shift Keying; 1 bit/symbol; Simple
FSK	Frequency Shift Keying; 1 bit/symbol; Modems
PSK/QAM	Phase (PSK) or Amp+Phase (QAM); Multiple bits/symbol; Wi-Fi, 4G

■ FINAL EXAM TIPS – Unit 1

Exam Strategy – Unit 1 ke liye:

1. ■ OSI Model – 7 layers table mandatory hai: name, function, protocol/example column ke saath
2. ■ Nyquist + Shannon dono numericals practice karo — SNR dB ko linear mein convert karna mat bhoolna
3. ■ Multiplexing mein FDM/TDM/WDM/STDM comparison table + ek line working zaroor likhna
4. ■ Circuit vs Packet Switching – 8+ feature comparison table format mein likhna
5. NRZ-L aur NRZ-I diagram: data 01001110 ke liye hath se draw karo — clearly label karo bits aur voltage levels
6. Guided media mein Twisted Pair (UTP/STP), Coaxial, Fiber – teeno ki bandwidth, distance aur use case yaad karo